FERA · BANGLA-FIRST CASE COORDINATION · SYNTHETIC PROTOTYPE

Reviewed Bangla reports. Officer decisions.

Reviewed intake, similarity-ranked candidates, and verified officer decisions.

Synthetic only · No real child data

[MOCK] officer verification

Referral recorded (simulation)

Draft outputs only

Hack for Humanity Bangladesh 2026 · Synthetic-data prototype · Built for review, not for delivery

WHAT FERA IS

A coordination workspace, not a notification system.

Coordinates

A parent reports a missing child by voice or typed Bangla.
Staff log an invented found-report description.
A verified officer compares ranked candidates and records a decision; counts are updated.

Output drafts (GD text, poster, summary) are generated from reviewed fields and labelled 'Draft — submit yourself'.

Does not

Send alerts to police or CID.
Auto-confirm, auto-close, or guess fields.
Take custody, return, or reunify children.
Store report payload server-side before the reporter confirms (metadata only is held).
Run synthetic provider processing without an on-screen notice (COMP-050 notice required).
Show candidates or other-party data to anyone except verified officers.

Sourced from AGENTS.md non-negotiables and D7 V2 release contract.

INTAKE · REVIEW BEFORE SUBMIT

Typed intake plus cached extraction [MOCK].

01

Start

Consent notice.
Bangla or English.

02

Speak or type

Voice: PCM WAV only.
WebM not supported.
Transcript editable.

03

Follow-up

Missing fields
asked one at a time
in Bangla.

04

Review

Every field shown
with its source span.
Unknown stays
unknown.

05

Confirm and submit

Stores the reviewed report; idempotent on client UUID.

Live AI provider availability is unverified. Cached outputs [MOCK] are labelled on screen and in narration.

Before submit: server holds only expiry, counter, hash and cost metadata; no report payload is stored (BR-004, FR-006).

Sources: D3 SCREEN-001–005; D1 BR-004, FR-001/004/006/026; AGENTS.md Principle I.

OUTPUTS · DRAFTS, NOT FILINGS

Manual review-before-submit, fixed templates, draft outputs.

GD draft text

Fixed template over reviewed fields. Label 'Draft — submit yourself / Draft — submit yourself' is always visible. Fera does not file a GD.

Poster preview

Fixed layout with synthetic fields only.
No photos, faces or exact locations.
Share and download are the parent's actions.

13219 summary

Brief, plain-language summary of the parent's own reviewed fields.
tel:13219 link provided; Fera does not place the call.

Sources: D3 SCREEN-004; D1 BR-002, BR-003; AGENTS.md Principle VI (Honest by construction).

OFFICER REVIEW · SIMILARITY, NOT IDENTITY

Staff entry, verified officer review, human decision.

Workflow

1. Staff log an invented found-child description (synthetic fields, no photos).
2. Matcher ranks candidates using filters, cosine similarity and a soft district bonus.
3. Verified officer opens the side-by-side evidence view and reads each source span.
4. Officer confirms or rejects with a reason.
5. System records a [MOCK] referral and writes an append-only audit row.

Reviewer checks

Score is 'similarity' — not identity probability.
Zero qualified candidates is a valid result.
Shared phone alone never establishes the same child.
Confirmation is one transaction: lock both rows, recheck versions, void competitors, write audit.
Candidates and other-party data stay scoped to verified officers (BR-024).

Sources: D3 SCREEN-007/009/010/011; D1 BR-001/012/013/017/018/024; AGENTS.md Principles II, III, IV.

EVALUATION · FROZEN METRICS NOT MEASURED YET

Each metric is 'not measured' until the live run.

Matching precision @k	*not measured — held-out run not executed*
Word error rate (WER)	*not measured — no reviewed 60-clip set*
End-to-end latency	*not measured — live latency not recorded*
Total spend (USD)	*not measured — no live paid calls*
Duplicate precision	*not measured — manifest not frozen; thresholds TBD*

Each metric is 'not measured' until the live run. Denominator and failures will be shown when measured (BR-026, Principle VI).

Sources: AGENTS.md Principle VI; D1 BR-026; D6 TASK-039–043.

WHAT WE BUILT · SYNTHETIC DATA

A working app, tested end to end on synthetic records.

In the repository

- 16 screens for parents, staff, officers and admins; Bangla and English.
- 22 API routes: Zod input, role guard, per-role DTO, error envelope.
- PostgreSQL 16 + pgvector in hand-written SQL; one-transaction confirm; append-only audit.
- Matcher: hard filters, multilingual-e5-small similarity, readable reasons.
- Bangla normaliser; LLM and speech adapters with labelled fallback.
- Duplicate detection; 64-district counts with an accessible table.

How it is checked

- 3,693 unit tests pass (308 skipped).
- 320 PostgreSQL + pgvector tests pass, fresh database per file.
- 78 Playwright + axe browser checks pass (desktop and mobile emulation).
- Strict TypeScript; lint 0 errors; remote CI green.
- Eleven specialist reviews, with a safeguarding gate on every PR.
- Software checks only: no human study or production run yet.

Team: Mahim (lead, platform) · Obidit (intake, AI, outputs) · Masnun (data, matching, officer workflow)

Counts: README verification at application commit 9155963 and remote CI; typecheck, lint and unit re-run 18 Oct 2026.

LIMITATIONS AND NEXT STEPS

What this prototype does not do, and what comes next.

Limitations

- Synthetic data only — no real child records, faces or coordinates.
- Officer verification is [MOCK]; referral is a recorded simulation.
- Live AI provider availability and account configuration are unverified.
- Voice intake supports PCM WAV only; webm is unsupported.
- Cached outputs [MOCK] are labelled and never hidden.
- No alert, runaway, rescue, police/CID-integration or legal-compliance claim.

Next steps

- Run the frozen held-out evaluation (TASK-043) and publish denominators.
- Human Bangla review of copy and a timed study with real participants.
- Configure a live AI provider within the USD 25 cap and deploy a hosted demo.
- Resolve OQ-011 (PDPA specifics) and OQ-018 (referral agency) before any pilot.
- Plan partner hand-off only after external identity and guardian procedures are confirmed.

Sources: D7 §4 clarifications still needed; PHASE_4_SUBMISSION.md AMD-4-1; AGENTS.md Principles VI, VII.